

Notation

I *Symbols* (Completely standard set-theoretic notation is not listed.)

<i>Symbol</i>	<i>Example</i>	<i>Use</i>	<i>Page</i>
juxtaposition	$s_p s_q, KL$	join	86
	K^r	r -skeleton of K	77
	fx	value of f at x	
	na	n integer, $a \in$ group	216
$\Rightarrow$		implies	
$\Leftrightarrow$		if and only if	
$\setminus$	$X \setminus Y$	difference of sets	
$\searrow^e$	$K \searrow^e L$	elementary collapse	89
$\searrow$	$K \searrow L$	collapse	89
$/$	B/A	quotient group	220
$ $	$\delta_1 \delta_2$	δ_1 is a factor of δ_2	231
$()$	(vw)	edge	11
	$(v^0 \dots v^p)$	p -simplex, p -chain	70, 100
	$K^{(r)}$	r th barycentric subdivision	135
$(,)$	(v, w)	edge	11
	(K, L)	pair	139
$(; ,)$	$(K; L_1, L_2)$	triad	159
$(, ,)$	(M, N, K)	triple	168
$\{ \}$	$\{z\}, \{z\}_K, \{z\}_{K,L}$	homology class	105, 114, 174
$\{ , \}$	$\{v, w\}$	edge	10
$\langle , \rangle$	$\langle \gamma, c \rangle$	value of cochain on chain	36
$\langle \rangle$	$\langle a \rangle$	subgroup generated by a	237

Symbol	Example	Use	Page
$\approx$	$M \approx N$	equivalence of closed surfaces	53
$\cong$	$A \cong B$	isomorphism of groups	217
$\#$	$M \# N$	connected sum	54
$\bullet$	$\overset{\bullet}{s}_n$	boundary of simplex	71, 76
$\circ$	$\overset{\circ}{s}_n$	interior of simplex	71
$<$	$s_p < s_n$	face of	72
$ \ $	$ K $	underlying space of	74
	$ A $	number of elements of	216
$-$	$\bar{s}_n$	closure of simplex	76
	$\bar{\Sigma}, \bar{\text{St}}$	closure of subset or star	84, 85
	$\bar{b}$	coset	220
	$\overline{M}$	closed surface obtained from surface	188
$\frown$	$(v^0 \dots \widehat{v^i} \dots v^p)$	simplex with vertices $v^0 \dots v^p$ except v^i	101
	$\widehat{\partial}$	relative boundary	113
	$\widehat{s}$	barycentre of	136
$\sim$	$z \sim z'$	homologous to	105
	$\widetilde{H}_0$	reduced homology group	104
	$\widetilde{i}_*$	reduced homomorphism	140
	$\widetilde{A}$	vector space from abelian group	233
	$\widetilde{f}$	linear map from homomorphism	234
$\sim(\text{mod } 2)$	$z \sim z'(\text{mod } 2)$	homologous mod 2 to	174
$\overset{L}{\sim}$	$z \overset{L}{\sim} z'$	homologous mod L	114
$\oplus$	$A \oplus B$	direct sum	222, 223
$\otimes$	$A \otimes B$	tensor product	179, 233
$'$	K'	barycentric subdivision	136
$''$	K''	second barycentric subdivision	183
$+$	V_i^+	region determined by V_i	183

II Roman characters

B	$B_p(K)$	boundary group	104, 124
	$B_p(K, L)$		114
	$B_p(K; 2)$		174
	$B_p(K, L; 2)$		176

<i>Symbol</i>	<i>Example</i>	<i>Use</i>	<i>Page</i>
C	$C_p(K)$	chain group	21, 99, 124
C	$C_p(K, L)$		113
	$C_p(K; 2)$		171
	$C_p(K, L; 2)$		176
	$C^p(G)$	cochain group	36
	CK	cone on K	145
Cl	$\text{Cl}(\Sigma, K)$	closure	84
F	FS	free abelian group on S	219
$\mathcal{F}$	$\mathcal{F}(R)$	frontier of region	30
$GL(n, \mathbb{R})$		general linear group	98
H	$H_p(K)$	homology group	104, 124
	$H_p(K, L)$		114
	$H_p(K; 2)$		174
	$H_p(K, L; 2)$		176
	$\tilde{H}_0(K), \tilde{H}_0(K; 2)$	reduced homology group	104, 174
$\mathcal{H}$	$\mathcal{H}(x)$	Heawood number	64
i_*		homomorphism from inclusion	121
$\tilde{i}_*$		reduced homomorphism	140
Im	$\text{Im} f$	image	104, 217
j, j_q		homomorphism forgetting a subcomplex	113
$\tilde{j}_*$		homomorphism from j	122
$\tilde{j}_*$		reduced homomorphism	140
Ker	$\text{Ker} f$	kernel	104, 217
Lk	$\text{Lk}(s, K)$	link	87
N		regular neighbourhood	183
P, kP		projective plane, k -fold	54
$\mathbb{Q}$		projective plane	
$\mathbb{R}^n$		rational numbers	218
		real euclidean or vector space of dimension n	
S		sphere	54
S^n		n -sphere	148
$\text{St}, \overline{\text{St}}$	$\text{St}(s, K)$	star, closed star	85
T, hT		torus, n -fold torus	54
Tor	$\text{Tor}(A, B)$	torsion product	179

Symbol	Example	Use	Page
V		subcomplex complementary to regular neighbourhood	183
w_p		Stiefel–Whitney class	178
Z	$Z_p(K)$	cycle group	104, 124
	$Z_p(K, L)$		114
	$Z_p(K; 2)$		174
	$Z_p(K, L; 2)$		176
$\mathbb{Z}$		additive group of integers	216
$\mathbb{Z}_k$		$\{0, 1, \dots, k - 1\}$ under addition modulo k	216
$\mathbb{Z}^n$		group of n -tuples of integers	218
III. Greek characters			
α, α_p		homomorphism	129, 167
	$\alpha_p(K), \alpha_p(K, L)$	number of p -simplexes of $K, K \setminus L$	55, 91, 155
β, β_p		homomorphism	130, 168
	$\beta_p(K)$	Betti number	155
	$\beta_p(K, L)$		155
	$\beta_p(K)$	connectivity number	174
γ, γ_p		homomorphism	168
δ		coboundary	36
∂, ∂_p		boundary homomorphism	21, 101, 172
∂	∂M	boundary of surface M	187
$\widehat{\partial}, \widehat{\partial}_p$		relative boundary	113
∂_*		homomorphism from ∂	123
$\widetilde{\partial}_*$		reduced homomorphism from ∂	140
Δ		homomorphism	159
	$\mathcal{G}^\Delta$	triangular graph	32
ε		augmentation	25, 103, 172
μ		cyclomatic number	19
ρ		restriction homomorphism	116
ϕ		homomorphism	158
χ	$\chi(K)$	Euler characteristic	55, 155
	$\chi(K, L)$		155
ψ		homomorphism	159